

Irmak Koseoglu

Machine Learning / AI Engineer

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SUMMARY

Master's student in Big Data and Artificial Intelligence, focused on **Generative AI** and **Machine Learning**. Experienced in building **Retrieval-Augmented Generation (RAG)** systems, training **deep learning** models, and deploying AI applications with **Python** and **PyTorch**. Motivated to build AI systems that work reliably in real products, not only in notebooks.

SKILLS

Languages & Frameworks: Python, C++, SQL, R

AI / Machine Learning: PyTorch, scikit-learn, TensorFlow/Keras, Hugging Face

LLM / Generative AI: LangChain, RAG, FAISS, ChromaDB, Prompt Engineering

Cloud & DevOps: Docker, Kubernetes, AWS, Microsoft Azure, GitHub Actions (CI/CD)

Data & Visualization: Pandas, NumPy, Matplotlib, Jupyter

Languages: English (fluent), Turkish (native), German (B1)

PROJECTS

PDF Question Answering API — *Retrieval-Augmented Generation Pipeline*

live demo

- Built and deployed a **RAG pipeline** that answers questions from PDF documents: text chunking, embeddings, **ChromaDB retrieval**, and LLM answer generation.
- Packaged as a **FastAPI** service with automated tests and a **CI/CD pipeline**; runs live in production.

Tech Stack: Python, FastAPI, LangChain, ChromaDB, Docker, GitHub Actions

LLM-Based Data Leakage Detection — *Benchmarking & Evaluation*

- Evaluated LLM capabilities for data leakage detection across 96 ML pipelines, comparing **prompt engineering strategies** against a rule-based baseline.
- Demonstrated that **Chain-of-Thought prompting** achieved the highest detection accuracy, uniquely identifying a subtle data leakage type.

Tech Stack: Python, OpenAI API, scikit-learn

Plant Disease Classification — *CNN vs. Transfer Learning*

- Trained a **ResNet50** model with **transfer learning** to classify 38 plant diseases from leaf images, significantly outperforming a baseline CNN.
- Implemented **Grad-CAM** visual explanations to verify model attention on leaf lesions, ensuring reliable predictions.

Tech Stack: PyTorch, ResNet50, Grad-CAM, FastAPI, Docker, GitHub Actions

Plant Disease Inference API — *MLOps Deployment*

- Deployed the plant disease model as a live **REST API** that serves top-k predictions with **confidence scores**.
- Built an automated **CI/CD pipeline** with **pytest** and **Docker** container builds on every push to ensure reliable deployment.

Tech Stack: FastAPI, Docker, GitHub Actions, pytest, PyTorch

Fake News Detection — *Text Classification*

- Preprocessed and vectorized a dataset of 72,000 news articles using **TF-IDF** and **feature engineering** to extract predictive text patterns.
- Evaluated multiple classification models and tuned hyperparameters with **GridSearchCV**, selecting the optimal model based on **accuracy** and **F1-score**.

Tech Stack: Python, scikit-learn

EDUCATION

SRH University of Applied Sciences Berlin

M.Sc. Big Data and Artificial Intelligence

Berlin, Germany
October 2025 – March 2027 (expected)

Istanbul Technical University (ITU)

B.Sc. Mathematics Engineering

Istanbul, Turkey
September 2019 – June 2024

EXPERIENCE

KPMG Turkey

Analyst, Digital Finance & Fintech

Istanbul, Turkey
October 2024 – June 2025

- Supported client projects on data and technology, working with cross-functional teams and reporting results clearly to both technical and non-technical stakeholders.